

EX. NO: 8 — OS NMAP LIVE HOST DISCOVERY

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Aim:

Discover live/active hosts in a network using scans such as ARP, ICMP, TCP and UDP.

Objective:

To understand how Nmap identifies active systems before performing further network scanning operations.

Software:

Nmap

Try hack Nmap from a computer with network access.

Theory:

Nmap is a free and open-source network scanning tool used to discover live hosts and identify network services. It was created by Gordon Lyon.

Nmap uses different live-host discovery methods using different protocols.

1) ARP Scan

Uses ARP scan requests to find hosts on the same subnet.

2) ICMP Scan

Uses ICMP requests and replies to identify hosts.

3) TCP Scan

Uses TCP requests and packets to specified ports and checks the response.

4) UDP Scan

Sends UDP packets and can identify an active host through an ICMP port unreachable response.

Observation:

- ARP request can discover hosts within the same subnet.
- ARP cannot cross routers because it operates at the link layer.
- An ACK ping generally requires a predefined port.
- UDP ping can identify an active host when an ICMP port unreachable is received.

- The -sn option allows Nmap to perform host discovery without conducting a port scan.

Result:

Thus, the live host discovery using Nmap was successfully studied and performed. The experiment helped in understanding how active hosts can be identified before performing further network scans.